

The Reader Is the Variable

Constraint Factorization, Free Moves, and Instrument Limits in Deterministic Transformation Fields

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Abstract

This paper audits a deterministic transformation field for structure that survives the removal of reader-side artifacts. Four channels previously treated as primitive — position, adjacency, value, and echo — reduce to two: value is the output of a read rule applied to structure, and an echo is that read rule failing to separate two structurally distinct states. The remaining primitives are position and adjacency, an unlabeled relational graph.

Two consequences are forced by this reduction and both correct previously stated architecture. First, a stateless reader cannot detect the convergence criterion the architecture specifies: periodicity is a property of a sequence, not of an output, so the reader must be a transducer carrying at least as much state as the return period. Second, under the corrected relational form of Co, return period is the only observable in the current vocabulary that is natively two-frame; every other candidate — band width, echo position, fiber size, value — is a single-frame reading requiring an externally supplied comparison.

The central computation is a factorization audit of the SHA-256 dependency lattice. The Davies–Meyer feed-forward separates the digest constraint into eight independent 32-bit conditions. That separation is destroyed by a single backward round: the ancestry of the last schedule word is the complete seed. Forward, the picture inverts — ten expanded words carry partial ancestry, with full coupling first reached at W[26]. The two factorizations do not meet. Under a single varying seed word the invariant cone contains exactly two expanded words, {16,17}, which coincides with a seam previously derived from an unrelated dual-rail criterion.

Four measured instrument failures are reported as constraints rather than discarded. Each shares one defect: a relational structure was reduced to a scalar before being tested for relational content. A canonical-section theorem is derived limiting any deterministic recovery procedure to at most one preimage per collision fiber, which sharpens the admissible-domain condition from smallness to sectionhood. Falsifiers are given for each surviving claim, including a base-scaling test that would reverse the four-to-two reduction.

Contents

1. The problem, stated without the conclusion.....	4
1.1 What is not claimed	4
1.2 Grading.....	4
2. The primitive reduction: four channels become two.....	5
2.1 Value is the output of a read rule	5
2.2 Echo is aliasing	5
2.3 The edge asymmetry is forced, not statistical.....	5
2.4 The echo perturbs the Sturmian staircase.....	6
2.5 Zero is a state, not an absence.....	6
2.6 What remains.....	7
3. Two forced corrections to the architecture	7
3.1 The reader cannot be a function.....	7
3.2 Return period is the only two-frame observable in the vocabulary.....	7
3.3 C2 forbids annihilation, not difficulty	8
4. The factorization audit	8
4.1 Setup.....	8
4.2 The feed-forward splits into eight.....	9
4.3 Backward: the split dies in one round.....	9
4.4 Forward: ten partially-coupled words	9
4.5 The free cone and an unexpected coincidence.....	10
4.6 What this means for an incremental procedure	10
5. Two kinds of free move, which must not be merged.....	11
5.1 Reader-free: the configuration moved and the reader aliased it.....	11
5.2 Generation-free: the relation was transported, not created	11
5.3 Why the two must stay separate.....	11
5.4 Free cones are a generation-free instrument	12
6. The canonical-section theorem	12
7. Four instrument failures, and the defect they share	13
7.1 Carry exhaust does not privilege the empty trajectory	13
7.2 The empty trajectory's apparent privilege is seed sparsity	14
7.3 Iterated hashing is a funnel, not a wheel	14
7.4 The horizon belongs to the instrument	15

7.5 The shared defect	16
7.6 How these nulls may be cited.....	16
8. The survivor/cancellation duality.....	17
8.1 J labels the split; S exchanges it	17
8.2 Applied to the corpus	17
8.3 The unnamed third generator.....	17
8.4 A retraction on generation	18
9. Falsifiers.....	18
9.1 Base scaling — the falsifier for the primitive reduction.....	18
9.2 Falsifiers for the remaining claims	18
10. Open questions, stated so they can be decided	19
10.1 Which reader.....	19
10.2 Does trajectory coherence survive scalar saturation	19
10.3 Does iteration produce stable relational orbits.....	19
10.4 The two-grid construction, relocated.....	19
10.5 Is the low-weight seed family informative.....	20
10.6 Does the front-end partial coupling support anything.....	20
11. Claim ledger	20
12. What the paper does and does not establish.....	21
Appendix A. Methods.....	23
A.1 Ancestry and free cone.....	23
A.2 Carry exhaust.....	23
A.3 Functional graph	23
A.4 Basin and controller	23
A.5 Reproducibility.....	23
Appendix B. Retracted claims retained as walls.....	23
Appendix C. Consistency audit	24

1. The problem, stated without the conclusion

A deterministic transformation is normally described as a function that consumes an input and emits an output. Under that description the input is gone and the only question about the output is what can be computed from it. This paper investigates a different description, and the difference is not rhetorical: it changes which objects are held fixed and which are allowed to vary.

Under the alternative description there is a fixed transformation grammar — constants, Boolean functions, rotations, modular arithmetic, schedule geometry, round topology, feed-forward structure — and there are states that interact with it. The grammar does not change between one input and the next. What changes is the trajectory taken through it. The output is then not a container but a boundary condition on that trajectory.

The question this paper asks is narrow and answerable: given the fixed grammar and a fixed output, does the constraint imposed by that output factor over the structure of the trajectory, or is it a single joint condition? The answer determines whether an incremental procedure has anything to work with. If the constraint factors, partial progress is possible and a feedback loop has handles. If it does not, every candidate is accepted or rejected whole, and no amount of sophistication in the loop changes that.

This is a question about a dependency lattice, not about cryptography, and it is computable. The answer, given in Section 4, is that the constraint factors at two ends and not in the middle, and that the two factorizations do not connect.

1.1 What is not claimed

SHA-256 remains deterministic, many-to-one over unrestricted inputs, avalanche-producing, and computationally one-way in the conventional sense. Nothing here overwrites those facts and nothing here constitutes an inversion, a preimage attack, a collision, or a reduction in the security of any deployed system. Section 6 derives a limitation that applies to any deterministic recovery procedure whatsoever, including every procedure suggested by this framework.

Nor is it claimed that a 256-bit digest contains an arbitrary message as recoverable stored information. For unrestricted inputs the compression function maps a larger domain to a smaller codomain and the pigeonhole principle settles the matter. The investigation concerns a conditional question about constrained domains, and Section 6 shows the relevant condition is stronger than the smallness usually assumed.

1.2 Grading

Every substantive claim in this paper carries one of the following grades, and the grade is stated at the point the claim is made. A consolidated table appears in Section 11.

Grade	Meaning
THEOREM	Follows deductively from stated premises; proof given or immediate.
DERIVED	Follows from the constraint system as stated, contingent on the constraints.
COMPUTED	Output of an exact finite computation; reproducible, no sampling.
MEASURED	Statistical result from finite sampling; effect size and dispersion given.

Grade	Meaning
FALSIFIED	A claim previously advanced in this program, now retracted, with the killing observation.
OPEN	Stated precisely enough to be decided; not decided here.

2. The primitive reduction: four channels become two

Earlier statements of this framework treated four channels as co-primitive: position P , adjacency or connectivity G , value V , and echo E . That inventory does not survive scrutiny, and the reduction that replaces it is the foundation for everything after.

2.1 Value is the output of a read rule

A hexadecimal string is not a number. It is an ordered sequence of symbols with adjacency relations. To obtain a number from it one must apply a rule: assign each symbol a weight, choose a base, choose a direction, and accumulate. Horner's recursion $x \rightarrow bx + d$ is such a rule. So is the choice of base. So is the decision to read left-to-right rather than right-to-left.

Therefore $V = \rho(P, G)$ for a reader ρ , and value is downstream of structure rather than beside it. This is not a philosophical preference; it is a statement about what a base conversion does. The same object under a different reader produces a different number, and no experiment can determine which number the object 'really' has, because the question is malformed.

The consequence for a digest is immediate. A 256-bit output is a fixed bit pattern. Its interpretation as an integer, as thirty-two bytes, as eight 32-bit words, as sixty-four nibbles, as a traversal seed, or as a set of coordinates is a choice of ρ . None of these is the object. [DERIVED]

2.2 Echo is aliasing

An echo was defined as a step in which the representation changes while the value does not. Written out:

$$(P, G) \neq (P', G') \quad \text{but} \quad \rho(P, G) = \rho(P', G')$$

That is the definition of aliasing. The structure moved; the reader did not resolve the move. So an echo is not an independent channel — it is a report on the reader's resolution, and it is doubly downstream, since it is defined in terms of V which is itself defined in terms of ρ . [DERIVED]

The concrete mechanism is worth stating because it is exact rather than statistical. Under left-edge lifting on a k -nibble string, the value transforms as $v \rightarrow v \bmod 16^{(k-1)}$. If the newly exposed leading nibble is zero, the represented integer is unchanged. The string is shorter, the head has moved, the adjacency structure has changed, and the number has not moved at all.

2.3 The edge asymmetry is forced, not statistical

Right-edge traversal removes the trailing nibble, giving $v \rightarrow \lfloor v/16 \rfloor$. For positive v this is strictly decreasing. Floor division has no fixed point above zero, so right-to-left traversal cannot produce an echo — not rarely, but never, for arithmetic reasons rather than distributional ones. [THEOREM]

Measurement over three thousand random 64-nibble strings confirms the asymmetry at the predicted magnitude. Left-to-right produced 3.955 echoes per full decay, a rate of 0.0628 per step against the predicted

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$1/16 = 0.0625$. Right-to-left produced 0.005 per decay. Center-out produced 0.009. The two edges differ by three orders of magnitude. [MEASURED]

This is the mechanism behind a visual observation recorded before the mechanism was known: left-to-right rendering of these decays shows horizontal banding, right-to-left shows vertical erosion with no repeated rows. The bands are the echoes. The right edge has no bands available to it because it has no fixed points available to it.

2.4 The echo perturbs the Sturmian staircase

The decimal length of a truncated hexadecimal string follows a Sturmian word with slope $\log_{10}16 = 1.2041199826559248$. The continued fraction of that slope is $[1; 4, 1, 8, 1, 9, 1, 25, \dots]$, and the convergent denominators are 4, 5, 44, 49, 485, 534. The value 49 is the largest denominator not exceeding 64, and $16^{49}/10^{59} = 1.0043363$, which is the closest a 64-nibble frame comes to a decimal boundary. The unusually large partial quotient 25 following that convergent is what makes it good. [COMPUTED]

The refinement this paper adds is that the staircase is not clean. On a sample decay the drop histogram was {0: 6, 1: 39, 2: 17, 3: 1}. The zero-drops are the echoes. The accounting closes exactly: $39 \cdot 1 + 17 \cdot 2 + 1 \cdot 3 = 76$, and 64 nibbles carry 77 decimal digits down to one, a total change of 76. The zeros steal steps from the staircase and force compensating larger drops elsewhere. [COMPUTED]

So the defect density in the Sturmian word is exactly $1/16$ and the defects sit at zero nibbles. Prior statements of this result described the word as clean ones and twos at the given slope and did not account for zero nibbles. [DERIVED]

2.5 Zero is a state, not an absence

A distinction that recurs throughout this work and was handled loosely in earlier drafts: 0 is not the empty set. Zero is a distinguished value produced by a reader. Absence of a relation is itself a relation. Representing nothing already introduces a distinction, so the representation is not the thing it purports to represent. [DERIVED]

This is not pedantry; it determines how every null in this paper may be read. Four statements that are routinely conflated:

zero observed value	≠	no transformation occurred
no distinction under p	≠	no distinction exists
empty fiber	≠	nothing
unresolved measurement	≠	absence of structure

The operative consequence is a reporting rule adopted throughout Section 7: no result in this paper says that a relation is not present. Every negative result says that under a stated transformation grammar and a stated reader, the relation did not render. The distinction is load-bearing because the entire method rests on the possibility that a reader can fail to resolve structure that is nonetheless there — which is precisely what an echo is. A framework that admits reader aliasing cannot also treat non-detection as nonexistence without contradicting itself.

The same discipline applies to the padded empty message. It is not the absence of an input. It is a specific seed of Hamming weight one, and Section 7.2 shows that this weight, not its emptiness, produces every anomaly it displays. Where earlier text in this program said 'the empty field,' the accurate phrase is either the fixed grammar or the weight-one seed trajectory, and these are different objects.

2.6 What remains

After the reduction the primitive structure is (P, G) : an unlabeled graph of positions and the adjacency relations among them. Value is a rendering. Echo is a degeneracy of that rendering. Both are reader-side. This is the reduction the rest of the paper depends on, and Section 9 gives the experiment that would reverse it.

3. Two forced corrections to the architecture

Both corrections in this section are repairs, not refinements. Each identifies a formulation that cannot do the job assigned to it.

3.1 The reader cannot be a function

The architecture specifies a reader as a map $\rho: (P, G) \rightarrow R$ and separately specifies the convergence criterion as a return of the configuration after r steps, with $r \geq 2$. These are incompatible.

A function is stateless. Given the same argument it returns the same value, and it has no access to the fact that it has been called before. Periodicity is not a property of an output; it is a property of a sequence of outputs. A stateless ρ evaluated at step n and again at step $n+r$ produces identical results in both the periodic case and the case where the configuration merely happened to repeat once, and it cannot distinguish either from a system that has been constant throughout. [THEOREM]

Therefore whatever registers the lock must carry state. The reader is a transducer:

$$\rho : (\Sigma, P, G) \rightarrow (\Sigma', R)$$

with $|\Sigma|$ at least the return period being tested. C2 — the requirement that a lawful continuation not sever the relational history producing the present state — was stated as a constraint on the field. The orbit criterion forces it onto the reader as well. A memoryless reader violates C2 by construction: it discards, at every step, exactly the history that would let it recognize a return.

This is why the earlier hill-climbing controller was structurally the wrong shape independent of its metric. It carried one state, the current best, and compared scalars. It had no capacity to notice that it had been somewhere before.

3.2 Return period is the only two-frame observable in the vocabulary

The corrected form of Co states that a distinction requires at least two non-identical relational frames. A single isolated state has no internal criterion by which it differs from anything. Applied to the measurement vocabulary this is unexpectedly restrictive.

Consider the candidates. A value is a single-frame reading. A band width is the size of one fiber. An echo position is one location in one trajectory. A fiber count is a property of one rendering. Each of these becomes a distinction only when a second frame is supplied from outside — a reference message, a control, a baseline — and the supply of that second frame is a choice made by the experimenter, not a property of the object.

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Return period is different. A period is a relation between a configuration and a later configuration. It cannot be computed from one frame, and it does not require an externally supplied comparison, because the second frame is the system's own later state. Under Co' it is the only quantity in the current vocabulary that is natively relational. [DERIVED]

This explains, rather than excuses, why the fixed-point criterion failed and the orbit criterion survived. A fixed point $Q(n+1) = Q(n)$ is a single configuration. Under Co' a single configuration renders no distinction, so a reader that arrives at a true fixed point has arrived somewhere it cannot read. Closure and reading are mutually exclusive. This is the unoccupiable seam arriving from the constraint side rather than by assertion, and it is the third independent route to that conclusion in this program.

The operational consequence: a tuner should not test whether motion has stopped. It should test whether the return period has stabilized, and it should test stability by perturbation — kick the configuration off the orbit and observe whether the same r is recovered. Both the quantity and the test are relational end to end, and nothing is summed.

3.3 C2 forbids annihilation, not difficulty

A third correction, weaker than the first two but consequential for how this paper's negative results may be cited. C2 states that a lawful continuation cannot sever the relational history producing the present state. It has sometimes been read as promising that history is explicitly stored and therefore retrievable. It promises neither.

The correct reading is narrower and harder to misuse: a lawful transformation cannot make relational ancestry unavailable in principle. The ancestry may be distributed across edges, grammar, constraints, alternate continuations, or a route so long that no procedure will traverse it. C2 forbids treating a lawful continuation as metaphysical annihilation. It says nothing whatsoever about cost. [DERIVED]

This cuts in both directions and the paper relies on both. Against the program: computational difficulty is fully compatible with C2, so the existence of a relation supplies no reason to expect a tractable route to it, and no result here may be read as evidence that inversion should be easy. Against the objection: the converse is equally true, and a demonstration that a route is expensive is not a demonstration that the relation has ceased to exist. Section 6 shows the actual obstruction is not cost at all but functionhood, which is a structural limit rather than a difficulty.

4. The factorization audit

This is the central computation. The question from Section 1 — does the output constraint factor over the trajectory structure — is a question about a dependency lattice and is decided by exact computation rather than sampling.

4.1 Setup

SHA-256 expands sixteen seed words into sixty-four schedule words by the recurrence

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}$$

and runs sixty-four rounds of an eight-word state, finishing with the Davies–Meyer feed-forward $H = S_{64} \boxplus IV$, componentwise modulo 2^{32} .

Define the ancestry $A(W[t]) \subseteq \{0, \dots, 15\}$ as the set of seed indices on which $W[t]$ transitively depends, with $A(W[t]) = \{t\}$ for $t < 16$ and $A(W[t]) = A(W[t-2]) \cup A(W[t-7]) \cup A(W[t-15]) \cup A(W[t-16])$ otherwise. Ancestry is a purely combinatorial object determined by the lag set $\{2, 7, 15, 16\}$; it does not depend on the constants, the Boolean functions, or the arithmetic.

4.2 The feed-forward splits into eight

The initialization vector is fixed and public, so $S_{64} = H \boxminus IV$ is recovered exactly from the digest. Modular addition couples bits within a word through carry propagation but never couples across words. Therefore the digest constraint is not one joint 256-bit condition and it is not 256 independent bit-conditions; it is exactly eight independent 32-bit conditions. [THEOREM]

The round function is likewise invertible given $W[t]$ and $K[t]$. Verified directly: a forward round applied to the initialization vector and then inverted returns the input state exactly. This is the standard observation that SHACAL-2 is a bijection for fixed key material, and it means that all irreversibility in the construction is located at the feed-forward. [COMPUTED]

4.3 Backward: the split dies in one round

Unwinding one round from S_{64} requires $W[63]$. Its ancestry is the complete seed:

$$|A(W[63])| = 16$$

So after a single backward step the constraint involves every seed word. The eight-way factorization present at the feed-forward does not survive one round of unwinding. The union of ancestries over consumed schedule words is already the full set at depth one and remains so at every greater depth. [COMPUTED]

This answers the load-bearing question in the negative for the backward direction. A tuner unwinding from the digest has one handle, not eight. The apparent eight-fold separation is real but is confined to a single step at the boundary, and it purchases nothing incremental.

4.4 Forward: ten partially-coupled words

The forward direction inverts the picture. The first expanded words carry narrow ancestry:

Word	A	Ancestry
$W[16]$	4	$\{0, 1, 9, 14\}$
$W[17]$	4	$\{1, 2, 10, 15\}$
$W[18]$	7	$\{0, 1, 2, 3, 9, 11, 14\}$
$W[19]$	7	$\{1, 2, 3, 4, 10, 12, 15\}$
$W[20]$	10	$\{0, 1, 2, 3, 4, 5, 9, 11, 13, 14\}$
$W[21]$	10	$\{1, 2, 3, 4, 5, 6, 10, 12, 14, 15\}$
$W[22]$	13	$\{0, 1, 2, 3, 4, 5, 6, 7, 9, 11, 13, 14, 15\}$
$W[23]$	14	$\{0, \dots, 10, 12, 14, 15\}$
$W[24]$	15	all but 12

Word	A	Ancestry
W[25]	15	all but 13
W[26]	16	complete

Full coupling is first reached at W[26]. Exactly ten expanded words — W[16] through W[25] — carry partial ancestry, and every word from W[26] to W[63] is fully coupled. [COMPUTED]

So the constraint factors at the seed end and at the digest end, and the two factorizations do not meet. Between W[26] and W[63] the system is globally coupled in both directions. Whatever incremental purchase exists is at the front of the schedule, not at the back, and the front is the end furthest from the digest. [DERIVED]

4.5 The free cone and an unexpected coincidence

The same lattice read in the other direction gives the free cone. For a set V of varying seed indices, a schedule word is provably invariant under every allowed variation when $A(W[t]) \cap V = \emptyset$. This is calculated before any expensive transformation is performed, not estimated afterward.

Take $V = \{3\}$, the case where a single seed word varies and the other fifteen are held. This is the structure of a constrained-input search: a counter, a timestamp, a nonce, or any protocol field that moves while a template stays fixed. The computation returns:

- First schedule word touching the varying index: W[18].
- Provably invariant expanded words: exactly two, W[16] and W[17].
- Rounds consuming only invariant schedule material: rounds 0, 1, 2.

The coincidence is that $\{16, 17\}$ is the same pair identified in earlier work in this program as the minimum sampling aperture, derived there from a dual-rail lattice criterion with no reference to input variation or free cones. Two independent derivations return the same pair. [COMPUTED]

The common cause is the lag geometry and nothing else. W[18] reaches index 3 by the lag-15 arm, since $18 - 15 = 3$, and W[16] and W[17] are the only expanded words whose four arms all land above index 3. The seam derivation and the free-cone derivation are two readings of the same lag structure, which is why they agree. This is a resonance worth recording, not an independent confirmation of either result. [INTERPRETATION]

4.6 What this means for an incremental procedure

The audit permits a precise statement of what a constraint-driven loop can and cannot have.

It can have: an exactly computed invariant cone for any specified variation, available before computation; an eight-way split at the digest boundary; a ten-word window of partial coupling at the seed end.

It cannot have: an incremental path from the digest backward, because the first step is already globally coupled; nor a connection between the front and back factorizations, because the middle is coupled in both directions.

This does not decide whether a relational reader operating on trajectory structure would find purchase where a constraint-propagation reader does not. It decides that constraint propagation from the digest is not the mechanism. [DERIVED]

5. Two kinds of free move, which must not be merged

Earlier drafts of this paper collected several phenomena under a single heading of free moves: the left-edge echo, transported dimension, invariant cones, cached prefixes. That collection was an error of exactly the kind this program exists to catch — a resemblance treated as an identity. There are two distinct phenomena and they differ in what is doing the work.

5.1 Reader-free: the configuration moved and the reader aliased it

A transformation is reader-free with respect to p when

$$X \neq T(X) \quad \text{but} \quad p(X) = p(T(X))$$

The structure genuinely changed. Position moved, adjacency changed, the configuration is not the one it was. The reader failed to separate the two. This is the aliasing class, and the left-edge echo is its cleanest instance: under $v \rightarrow v \bmod 16^{(k-1)}$ with a zero in the exposed position, the string shortens, the head advances, and the rendered integer does not move. [DERIVED]

The cost is zero in the value channel and non-zero in the structure. Nothing was saved; something was simply not seen. Whether this is exploitable depends entirely on whether the thing being computed is the rendering or the structure, and confusing the two is how a reader-free move gets mistaken for a saving.

5.2 Generation-free: the relation was transported, not created

A component is generation-free at a location when it is inherited from elsewhere rather than newly produced there. Nothing is aliased and no reader is involved. The claim is about provenance.

The round function supplies an exact instance. Written out, one SHA-256 round maps

$$(a, b, c, d, e, f, g, h) \rightarrow (T1+T2, a, b, c, d+T1, e, f, g)$$

Six of the eight output words are input words relabelled. No arithmetic touches them; they change position and nothing else. Two words, a and e , are computed. The generation-free fraction per round is exactly $6/8 = 0.750$, and across the compression each word is created once and transported seven times before reaching the feed-forward. [THEOREM]

This is not the framework's language applied to SHA. It is what the round does, and the ratio is a design constant rather than a measurement. The most widely deployed hash function in the world has a round that is three-quarters transport by construction.

5.3 Why the two must stay separate

The distinction is not terminological. The two classes differ in every property that matters.

	Reader-free	Generation-free
What changed	The structure did	Nothing, at this location
Why it is invisible	ρ lacks resolution	There is nothing new to see
Depends on the reader	Yes, entirely	No
Removed by a finer reader	Yes	No — it is not a resolution effect
Can be exploited to skip work	Only if the rendering is the target	Yes, unconditionally
Instance	Left-edge echo	Round relabelling; invariant cones

The consequence is that a finer reader dissolves the first class and leaves the second untouched. Merging them would predict that improving resolution eliminates the round's transport structure, which is false. This is the specific false identification the earlier draft committed, and it is corrected here. [DERIVED]

5.4 Free cones are a generation-free instrument

The lineage computation of Section 4.5 belongs to the second class. It identifies material that cannot have been affected by a specified variation — provenance, not resolution — and it therefore holds under every reader. Given a grammar with known dependency structure and a set V of varying inputs, one topological pass suffices:

```

for each node Y in topological order:
  A(Y) = union of A(inputs of Y)
  if A(Y)  $\cap$  V =  $\emptyset$ : mark GENERATION-FREE
  else: mark ACTIVE
return: generation-free set, active set, boundary  $t^*$ 

```

The number t^* answers a question useful independent of any inversion program: how long the machine can run before the varying thing can possibly matter. It is calculated rather than benchmarked, it is exact, and because it is generation-free it is not contingent on any choice of rendering. [DERIVED]

6. The canonical-section theorem

This section derives a limitation on any deterministic recovery procedure, including every procedure this framework might propose. It is stated here rather than buried because it sharpens the admissible-domain condition in a way that matters.

Theorem. Let κ be any deterministic procedure consuming only a digest H and the fixed grammar F , and returning a candidate message. Then κ recovers at most one member of each collision fiber of the hash, regardless of the reader, the number of iterations, the coherence functional, or the internal architecture of the loop.

Proof. κ is a function of H , since F is fixed and the procedure is deterministic. A function has exactly one output per input. All members of a collision fiber share the same H . Therefore all of them map to the same output under κ , and at most one of them equals it. ■ [THEOREM]

The consequence is that the admissible domain $\mathcal{A}$ in any conditional recovery claim cannot merely be small. It must be a partial section of the hash: at most one member per fiber. These are different conditions, and the difference is load-bearing.

Short messages are usually offered as the canonical example of a workable restricted domain. But shortness is not sectionhood. Short messages form a partial section only if no two of them collide — which is believed but not proven, and is exactly a cryptographic hardness assumption. Any claim of the form 'recovery is possible on restricted domains' is therefore resting on that assumption, and honest presentation requires naming it rather than letting shortness stand in for it. [DERIVED]

This is the correct location for the information-theoretic boundary in the argument. It is not an external objection imported to deflate the program. It follows from determinism, which is the property the program relies on most heavily, and it specifies the target precisely rather than forbidding it.

7. Four instrument failures, and the defect they share

Negative results are reported here as constraints on future instruments rather than discarded. All four share a single defect, identified in Section 7.5, and identifying it is more valuable than any of the individual nulls.

7.1 Carry exhaust does not privilege the empty trajectory

Construction: Track A runs the real schedule with modular addition; Track B runs the identical recurrence with XOR, giving the carry-free linear shadow $W = L \cdot M$. The carry exhaust is $\Delta = A \oplus B$, isolating exactly the bit positions where carry propagation changed the result. Bits were encoded as ± 1 in $\mathbb{R}^{2048}$ so that projection is well-posed; over $\text{GF}(2)$ the projection formula divides by a quantity that can vanish for nonzero vectors.

Three arms were run, projecting each message trajectory against a reference and measuring the residual:

Arm	Reference	Residual norm	Correlation
1	Empty trajectory	0.968	+0.24913
2	Unrelated message	0.968	+0.25054
4	Weight-matched sparse control	0.968	+0.25042

All three arms agree to three decimals. The empty trajectory explains no more of a message trajectory than an unrelated message does. [MEASURED]

The shared component is fully accounted for. The per-position mean μ , estimated from six hundred training messages, satisfies $\|\mu\|^2/2048 = 0.25306$, which predicts the observed correlation to three decimals. Centering by μ collapses everything: message-to-message +0.00088 with dispersion 0.0247, empty-to-message +0.00020, sparse-to-message +0.00121, against a chance dispersion of 0.0221. Five hundred fourteen of the 2048 bit positions — eight per word — were identical across all six hundred training messages. [MEASURED]

Interpretation: a large, highly significant, entirely message-independent structure exists in the carry channel. It is the beat between the linear schedule and the carry automaton, and it belongs to the field. A detector pointed at it confirms the grammar and recovers nothing about the query. [DERIVED]

7.2 The empty trajectory's apparent privilege is seed sparsity

The empty padded block has seed Hamming weight one. Random padded seeds average 129. Carries require bits to propagate, so the empty trajectory is anomalously sparse for reasons unrelated to emptiness.

Trajectory	Seed weight	Carry-exhaust weight (of 2048)
Empty message	1	514
Weight-matched sparse control	3	560
Random messages	~129	768 ± 21

Both low-weight seeds sit roughly twelve standard deviations below the random distribution, and the positional profile shows why: zero carry bits at every schedule position from $t = 16$ through $t = 24$ for both. After centering, the empty trajectory and the weight-matched control correlate with each other at $+0.248$ while each sits at chance against random messages. [MEASURED]

Sparsity is the variable; emptiness is not. Any future claim that the empty trajectory is a privileged reference must control for seed weight, or it is measuring the wrong thing. Note also that this is a real class — low-weight seeds form a family with shared structure — and it was reported here as a confound to be removed. Whether the family is itself informative is open.

7.3 Iterated hashing is a funnel, not a wheel

The proposal that repeated hashing rotates a digest on a wheel is decidable at reduced width by exhaustive computation of the functional graph.

Quantity	Width 16	Width 20	Random function	Bijection
Image after 1 iterate	0.6314 N	0.6319 N	0.63212 N	1.0000 N
Image after 2 iterates	0.4684 N	0.4681 N	0.46850 N	1.0000 N
Image after 3 iterates	0.3736 N	0.3736 N	0.37403 N	1.0000 N
Cyclic nodes	551	1650	321 / 1283	N

The measured image fractions match the random-function values to four decimal places at both widths. A rotation is a bijection and would give unit image fraction at every iterate with every node cyclic. [COMPUTED]

Independently, the displacement from one iterate to the next was tested for reproducibility across seeds, since a wheel steps the same way every time. The angle between successive iterates measured $89.893^\circ \pm 3.667^\circ$ against a chance expectation of $90.000^\circ \pm 3.581^\circ$, and the displacement vector correlation across five

hundred distinct seeds measured -0.00335 ± 0.0641 against a chance dispersion of 0.0625. Orthogonal every time, in a fresh plane every time. A wheel has a rotation axis; this has no fixed plane. [MEASURED]

A separate and simpler argument closes the two-known-sides construction. The second iterate is a function of the first, so the preimage set $\{M : \text{SHA}(M) = H_1\}$ is identical to $\{M : \text{SHA}(M) = H_1 \text{ and } \text{SHA}(\text{SHA}(M)) = H_2\}$. Double hashing adds no constraint on M. This is a set identity, not a measurement. [THEOREM]

What double hashing does accomplish is worth stating because it is structural. SHA-256 is a Merkle–Damgård construction, so a digest is a resumable chaining state and length extension is possible. Folding a second time makes the output non-continuable. In the framework's own terms, a construction has been built specifically to produce a state that cannot be continued from — C1 violated by engineering, at the interface, on purpose. [INTERPRETATION]

7.4 The horizon belongs to the instrument

A local Hamming-distance controller was measured against reduced-round SHA-256. Two probes were run. The first, comparing random messages, returned noise, and did so because random 512-bit messages all sit near maximal distance from one another — a global probe of a local question. The corrected probe measures perturbations of the true preimage.

Rounds	k=1	k=2	k=4	k=8	k=16	k=256	Spread
2	2.0	4.8	6.8	12.3	25.2	63.7	+61.7
4	9.5	23.0	37.4	52.3	76.9	128.6	+119.1
6	30.6	45.1	74.1	103.9	121.0	127.5	+96.8
8	33.6	70.2	96.5	122.8	127.4	127.8	+94.2
12	74.4	106.9	120.6	128.0	127.6	128.4	+54.1
16	106.6	124.4	128.4	127.4	128.7	128.5	+21.9
20	127.4	126.7	127.6	127.6	127.4	127.2	-0.2
24	128.0	127.8	128.2	127.8	128.3	127.6	-0.5
64	129.5	129.1	126.7	126.9	127.5	127.9	-1.6

The boundary sits between sixteen and twenty rounds. At sixteen a single flipped bit still moves the output only 106.6 bits instead of the saturated 128; at twenty it is indistinguishable from an unrelated message. [MEASURED]

A closed-loop single-bit hill climber, run against equal-budget random search, locates the same boundary independently: it wins at rounds 6, 8, 10, 12, 14, and 16 by a consistent five to six bits (93.2 vs 97.8; 93.1 vs 99.0; 92.4 vs 99.1; 92.9 vs 98.1; 92.4 vs 99.4; 93.8 vs 99.5) and then reverses, losing at twenty (102.9 vs 99.5) and twenty-four (102.8 vs 97.1). [MEASURED]

The reversal is a diagnostic rather than a decay. Above the boundary local information has not gone absent; it has gone misleading, and a controller attending to it does worse than one ignoring it. The mechanism is

visible in the row shape at sixteen rounds, where $k = 1$ gives 106.6 but $k = 2$ already gives 124.4: the surviving basin has narrowed to a single bit, which is exactly the granularity of the controller's move set.

Retraction. This paper previously asserted that ~17 rounds is the maximum horizon for any local controller, on the grounds that no move is finer than one bit. That does not follow. The argument establishes a bound for controllers navigating by the measured metric on the measured representation. A controller navigating by phase, recurrence, return period, or trajectory coherence is not constrained by a Hamming basin, and no measurement here bears on it. The horizon belongs to the instrument. [FALSIFIED]

7.5 The shared defect

Every failed detector in this program reduced a relation to a scalar before reading it. The carry-exhaust arms took norms. The orbit test took Hamming distances. The basin probe took mean distances. The rejected cross-spectrum would have multiplied two transforms and summed. In each case the reduction was the step that determined the answer.

Summation is precisely the operation that destroys path information: it maps many orderings to one total. A detector that sums before comparing is asking whether two sums agree, which is a question about sums. The nulls obtained are therefore nulls about scalar reductions of these trajectories, and they do not generalize to the trajectories themselves. [DERIVED]

The corresponding methodological rule, stated for future work: the interaction must be formed at the structural level and read afterward, never the reverse.

WRONG: structure $\rightarrow$ scalar $\rightarrow$ compare

RIGHT: $(P_1, G_1), (P_2, G_2) \rightarrow$ interaction $\rightarrow$ read

Stated as an ordering law: transform first, preserve the transformation history, render second, compare renderings third, and calculate only if the calculation preserves the relation being tested. Never begin with a sum over trajectory elements when the object of investigation is the relation among them, because summation maps many orderings onto one total and that is the information being sought. [DERIVED]

7.6 How these nulls may be cited

Under the discipline of Section 2.5, none of the four results above says a relation is absent. Each says a relation did not render under a stated grammar and a stated reader. Written correctly:

- The carry channel did not discriminate messages under the $\pm_1$ encoding with scalar projection after centering by μ . It is not established that the carry channel carries no message-specific structure under any rendering.
- The weight-one seed trajectory was not privileged as a projection reference against weight-matched controls. It is not established that it is unprivileged under a structural reader.
- Iterated hashing is a contraction rather than a bijection. This one is stronger than the others — it is exact and reader-independent, since image cardinality is not a rendering. The wheel hypothesis is dead in every representation.

- Local Hamming structure saturates between sixteen and twenty rounds. It is not established that trajectory structure saturates there, and Section 7.4 retracts the claim that it does.

The third item is the only one of the four that generalizes, and it generalizes because it was not a scalar reduction. Counting the image of a map is a statement about the map.

8. The survivor/cancellation duality

The framework decomposes a transformation's output into a surviving component R and a cancelling component E , and asks what is invariant when the two are exchanged. The operator usually invoked for that exchange is the wrong one, and correcting it produces a selection rule rather than a symmetry.

8.1 J labels the split; S exchanges it

The involution J acts as $+1$ on the residue and -1 on the exhaust. Its eigenspaces are R and E . It labels the decomposition; it does not exchange the halves. An operator S with $S(E) = R$ and $S(R) = E$ is an isomorphism $E \rightarrow R$, and therefore exists only when the two have equal dimension.

$$\dim E = \dim R \quad \Leftrightarrow \quad \text{the decomposition is flippable}$$

This is a selection rule. The duality treated as universally available is a conditional structure that only balanced decompositions admit, and the right question is not what survives the flip but which decompositions are flippable at all. [DERIVED]

8.2 Applied to the corpus

The Davies–Meyer feed-forward consumes a 256-bit pre-feedforward state and a 256-bit chaining value and emits 256 bits. Here $\dim E = \dim R = 256$ exactly. The single engineered forget in SHA-256 sits precisely at balance — not approximately, exactly — which is why it is one clean spill rather than a smear. [DERIVED]

The Tusi couple, with a one-dimensional axial residue against a one-dimensional transverse exhaust, is likewise balanced, which is why it is a mechanism and not a discovery: rotate the frame and residue and exhaust trade places. It is pure coordinate.

A type III_1 algebra against a one-parameter modular flow is not balanced, and no S exists. This reframes the crossed-product construction: it is not repairing a pathology but manufacturing the missing balance by adjoining exactly the dimension that was absent. Takesaki duality — crossing by the modular group and then by its dual returns the stabilized algebra — makes the construction an involution. The crossed product is S . The exchange operator this framework has been seeking was already present in its own corpus, labelled as a repair. [INTERPRETATION]

8.3 The unnamed third generator

With $S^2 = 1$, $J^2 = 1$, and S conjugating J to $-J$, the two anticommute: $JS = -SJ$. These are the relations of σ_z and σ_x , and they generate an $\text{su}(2)$ with a third element iJS that neither the survival reading nor the cancellation reading names. Two poles are labelled and the equator is empty. [DERIVED]

The framework's non-zero lean ΔL is a candidate for that generator, and the identification would convert an asserted geometric necessity into an algebraic one: σ_y shares no eigenvector with σ_z , so a state diagonal in

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the survivor/exhaust basis cannot have definite lean, and the lean cannot be zero. This is a structural hypothesis, not a theorem, because the identification of ΔL with iJS has not been established. [OPEN]

8.4 A retraction on generation

The balance criterion was applied to a certified closure table and used to claim that generating frames are exactly the balanced non-degenerate ones, and further to retract the uniqueness of a particular frame's generative behaviour. That application is withdrawn here. The spaces R , T , and E in the closure table are dimensions of relation lattices over a number field; the spaces R and E in the survivor/exhaust decomposition are components of a transformation output. No map between them has been constructed, and applying a criterion derived in one setting to a table computed in another is exactly the resemblance-for-identity error this program is meant to avoid. The closure table stands; the balance reading of it does not. [FALSIFIED]

9. Falsifiers

Each surviving claim is paired with an observation that would kill it. A claim without such a pairing is not admitted to the ledger.

9.1 Base scaling — the falsifier for the primitive reduction

This is the most important falsifier in the paper because it can reverse Section 2, on which everything else rests.

If a band is a fiber of the value reader, then band statistics must scale with reader resolution. The left-edge echo rate is $1/b$ in base b , forced by the mechanism rather than measured. Render the same underlying object in base 4 and base 256 and the fiber density must move to $1/4$ and $1/256$ respectively, with band width and split geometry tracking accordingly.

- If band statistics track $1/b$: bands are fibers of ρ , and the four-to-two reduction stands.
- If band geometry is invariant under base change: bands are not reader artifacts but properties of (P,G) itself, the reduction fails, and the primitive layer is (P, G, band) .

The second outcome would undo the central move of this paper. It is stated here without hedging because a reduction that cannot be reversed by evidence is not a result. [OPEN]

9.2 Falsifiers for the remaining claims

Claim	Would be falsified by
Reader must be a transducer	A stateless construction that provably detects period $r \geq 2$ from single outputs.
Return period is the only two-frame observable	A single-frame quantity that establishes a distinction without an externally supplied comparison.
Backward factorization dies in one round	Any lag set variant where $ A(W[63]) < 16$; the claim is specific to $\{2,7,15,16\}$.

Claim	Would be falsified by
Free cone under $V=\{3\}$ is $\{16,17\}$	Direct recomputation; the result is exact and combinatorial.
Round is 6/8 relocation	Inspection of the round function; the claim is definitional.
Canonical-section theorem	A non-deterministic or side-information-consuming procedure — which is outside the theorem's scope, not a counterexample.
Carry channel is message-independent	A detector on the carry channel that discriminates messages after centering by μ .
Iterated hashing is a contraction	Image fraction at unity across iterates at any width.

10. Open questions, stated so they can be decided

10.1 Which reader

The candidate list — left-edge traversal, right-edge, center-out, step-k, gap sequence, echo positions, decimal staircase, lineage traversal, register path, schedule adjacency, iterated-hash orbit — has no ordering, and none is preferred here. A reader earns priority by producing reproducible structural discrimination against matched controls, not by rhetorical fit. This paper deliberately does not choose one, because choosing one and calling the choice a test is the failure mode of Section 7.

10.2 Does trajectory coherence survive scalar saturation

Two signals can each be individually indistinguishable from noise while remaining phase-locked. The measurements in Section 7 constrain sums and say nothing about this. The question is open in both directions: nothing here establishes that coherence exists, and nothing here establishes that it does not. Stating it as open rather than as either a promise or a null is the accurate position. [OPEN]

10.3 Does iteration produce stable relational orbits

Rendered under a structural reader rather than a bit vector, do hash iterates exhibit stable return periods, and are those periods recovered after perturbation? This is the only convergence criterion Section 3 permits, and it has not been tested. [OPEN]

10.4 The two-grid construction, relocated

The carry-channel version of the two-grid hypothesis is dead and Section 7.1 keeps it dead. It should not be rescued by redefining its test. But the construction was misidentified rather than refuted: it compared two message trajectories as values, which is the scalar-reduction defect of Section 7.5, not a two-grid experiment.

The native form applies two readers to one structure. Given two readers on the same (P,G) , each induces its own fiber decomposition. Bands under the first and bands under the second need not coincide. The observable is the relation between the two degeneracy structures — where they hold together, where one splits while the other does not, and how the boundary between agreement and disagreement moves under transformation. Neither reader alone contains this; it exists only in the comparison, which is what a beat is. [OPEN]

Two conditions must be met before such a measurement means anything, and the carry version violated both. The interaction must be formed at the fiber level and read afterward, never summed first. And it must discriminate against matched controls, since Section 7.1 demonstrated that a large, coherent, highly significant interaction pattern can belong entirely to the shared grammar. A beat is not evidence until it separates queries.

10.5 Is the low-weight seed family informative

Section 7.2 established that low-weight seeds form a real class with shared carry structure, and treated that class as a confound. Whether the class carries structure of its own, and whether it constitutes a usable restricted domain in the sense of Section 6, has not been examined. [OPEN]

10.6 Does the front-end partial coupling support anything

Section 4.4 found ten expanded words with partial ancestry at the seed end. Whether that window supports incremental constraint propagation in the forward direction — the opposite direction from the one usually attempted — is not decided by the ancestry computation alone, since narrow ancestry is necessary but not sufficient for a usable handle. [OPEN]

11. Claim ledger

Every substantive claim in this paper, with its grade. Falsified entries are retained deliberately: a killed claim marks a wall, and a wall indicates a direction.

§	Claim	Grade
2.1	Value is the output of a read rule; $V = p(P, G)$	DERIVED
2.2	Echo is aliasing, doubly downstream of p	DERIVED
2.3	Right-edge traversal cannot echo (floor division has no fixed point)	THEOREM
2.3	Echo rate 0.0628/step left, 0.005/decay right, 0.009 center	MEASURED
2.4	Convergent denominators of $\log_{10} 16$ are 4, 5, 44, 49, 485, 534	COMPUTED
2.4	Sturmian defect density is exactly 1/16, defects at zero nibbles	COMPUTED
2.5	Zero is a state, not an absence; non-detection is not nonexistence	DERIVED
3.1	A stateless reader cannot detect period $r \geq 2$	THEOREM
3.1	The reader must be a transducer; C_2 applies to it	DERIVED
3.2	Return period is the only natively two-frame observable available	DERIVED
3.3	C_2 forbids annihilation, not cost	DERIVED
3.2	Closure and reading are mutually exclusive under Co'	DERIVED
4.2	Feed-forward gives exactly 8 independent 32-bit conditions	THEOREM
4.2	Round is invertible given $W[t]$, $K[t]$ (verified directly)	COMPUTED

§	Claim	Grade
4.3	$ A(W[63]) = 16$: backward factorization dies in one round	COMPUTED
4.4	Ten partially-coupled expanded words; full coupling at $W[26]$	COMPUTED
4.4	Front and back factorizations do not meet	DERIVED
4.5	Free cone under $V=\{3\}$ is exactly $\{16,17\}$	COMPUTED
4.5	$\{16,17\}$ coincides with the independently derived seam	INTERPRETATION
5.2	Round is exactly $6/8$ transport, $2/8$ creation	THEOREM
5.3	Reader-free and generation-free are distinct classes	DERIVED
5.3	Merging the two free-move classes (earlier draft)	FALSIFIED
6	κ recovers at most one member per collision fiber	THEOREM
6	Admissible domain must be a partial section, not merely small	DERIVED
7.1	Three arms identical at 0.968; empty not privileged	MEASURED
7.1	$\ \mu\ ^2/2048 = 0.25306$ accounts for all shared structure	MEASURED
7.2	Empty-trajectory anomaly is seed sparsity, not emptiness	MEASURED
7.3	Iterated hashing matches a random function, not a bijection	COMPUTED
7.3	Double hashing adds no constraint on M (set identity)	THEOREM
7.3	Double hashing deliberately destroys continuation	INTERPRETATION
7.4	Hamming basin narrows to one bit at $R=16$, gone at $R=20$	MEASURED
7.4	Tuner beats random at $R=6-16$, loses at $R=20-24$	MEASURED
7.4	~ 17 rounds bounds every possible local controller	FALSIFIED
7.5	All four nulls share the scalar-reduction defect	DERIVED
7.6	Only the contraction result generalizes across readers	DERIVED
10.4	Two-grid is two readers on one structure, not two trajectories	OPEN
8.1	$S \neq J$; exchange requires $\dim E = \dim R$	DERIVED
8.2	Davies–Meyer sits at exact balance, $256 = 256$	DERIVED
8.2	The crossed product is S	INTERPRETATION
8.3	$JS = -SJ$ generates $\mathfrak{su}(2)$ with an unnamed third element	DERIVED
8.3	ΔL is that third generator	OPEN
8.4	Balance criterion determines which BBP frames generate	FALSIFIED
9.1	Band statistics must scale as $1/b$ if bands are fibers	OPEN

12. What the paper does and does not establish

Established. The primitive layer of this framework is (P, G) , with value and echo as reader-side outputs, subject to the base-scaling falsifier. A stateless reader cannot detect the framework's own convergence criterion.

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Return period is the only natively relational observable in the current vocabulary. The digest constraint separates into eight independent conditions at the feed-forward and is globally coupled after one backward round. Ten expanded words carry partial ancestry at the seed end, with full coupling at $W[26]$. Under single-word variation the invariant cone is exactly $\{16,17\}$. The round function is three-quarters relocation. Any deterministic recovery procedure is a partial section on collision fibers.

Not established. That any reader finds trajectory structure where scalar readers found none. That iterated hashing produces stable relational orbits under any rendering. That the digest–field interaction converges to anything. That the front-end partial coupling supports incremental propagation. That the lean is the third $su(2)$ generator.

Withdrawn. That the Hamming horizon bounds every possible controller. That the balance criterion governs which BBP frames generate. That reader-free and generation-free moves are one phenomenon — an earlier draft of this paper collected both under a single heading, and they differ in every property that matters, most sharply in that a finer reader dissolves the first and leaves the second untouched.

The honest summary is that the constraint-propagation route from the digest backward is closed by the ancestry computation, and that this closure is a genuine result rather than a discouragement: it removes a large branch of the search space and relocates the remaining question to the front of the schedule and to readers that do not sum. Whether anything survives there is open, and the instruments capable of deciding it have not yet been built — because every instrument built so far reduced the relation to a scalar before asking whether the relation carried information.

Appendix A. Methods

A.1 Ancestry and free cone

Exact set computation over the lag recurrence, no sampling. $A(W[t]) = \{t\}$ for $t < 16$; otherwise the union of ancestries at lags 2, 7, 15, 16. Free cone: mark $W[t]$ free when $A(W[t]) \cap V = \emptyset$. Round inversion verified by applying a forward round to the initialization vector with an arbitrary schedule word and inverting exactly.

A.2 Carry exhaust

Track A: real schedule, modular addition. Track B: identical recurrence with XOR, seeded identically, using its own history — the full linear shadow, not a per-step carry. $\Delta = A \oplus B$ per word. Bits encoded as ± 1 in $\mathbb{R}^{2048}$. Reference mean μ estimated on 600 held-out random messages; 400 test messages; scalar projection residual normalized by $\sqrt{2048}$.

A.3 Functional graph

Truncated map $g(x)$ = first n bits of SHA-256 of x encoded as four big-endian bytes, at $n = 16$ and $n = 20$. Full graph enumerated; image sizes by unique count under iteration; cyclic nodes by tortoise-free path marking. Random-function reference values from $t(n+1) = \exp(-(1 - t(n)))$, $t(0) = 0$, image fraction $1 - t(n)$.

A.4 Basin and controller

Reduced-round compression: schedule expanded to R words, R rounds from the initialization vector, feed-forward applied. Basin probe: 120 trials per cell, X obtained by flipping k of 512 seed bits of a known preimage. Controller: greedy single-bit-flip sweeps over a random permutation of 512 positions, budget 4000 evaluations, eight independent runs per round count; random-search control uses the same budget in independent draws.

A.5 Reproducibility

All figures in this paper are outputs of finite deterministic programs with fixed seeds where sampling is involved. Sampled quantities are reported with dispersion and against explicit chance baselines. Exact quantities are marked COMPUTED and involve no sampling.

Appendix B. Retracted claims retained as walls

This program keeps killed claims in the record because a wall indicates a direction. The following were advanced in the course of this work and are withdrawn.

- The Hamming horizon of ~17 rounds bounds every possible local controller. Killed by scope: the argument constrains controllers navigating by the measured metric on the measured representation only.
- The balance criterion $\dim E = \dim R$ determines which BBP frames generate. Killed by category: no map was constructed between the closure-table dimensions and the survivor/exhaust decomposition.

- The empty trajectory is a privileged reference in the carry channel. Killed by the weight-matched control.
- Iterated hashing is a rotation. Killed by the functional-graph image fractions and by the non-reproducibility of the displacement.
- The digest and its second iterate constitute two independent measurements. Killed by a set identity.
- The cross-spectrum of two trajectories against shared machinery is a test for message structure. Killed before running: it returns a large coherent result regardless of whether structure exists.
- Reader-free and generation-free moves are the same phenomenon. Killed by the resolution test: a finer reader removes aliasing and cannot remove transport.
- A scalar null is a structural null. Killed by the framework's own admission of reader aliasing, which makes non-detection and nonexistence formally distinct.

Appendix C. Consistency audit

Before release this paper was passed through a mechanical audit of its own prose, not a reading. Every string in the source was extracted and tested against five checks derived from the distinctions the paper itself insists on. The audit is reported because a paper that claims non-detection is not nonexistence should be able to demonstrate that it obeyed its own rule rather than assert that it did.

Check	Question	Result
A	Is every use of 'free' labelled reader-free or generation-free?	4 flagged, all false positives (carry-free, free-cone, tortoise-free, pseudocode inside §5.4). Pseudocode relabelled for zero ambiguity.
B	Does any null assert absence rather than non-rendering?	1 flagged; on inspection it is the correctly-formed statement that carry structure is not established absent under any rendering. No violations.
C	Does any C2 claim imply computational accessibility?	5 C2 mentions; 1 flagged, on inspection it states that difficulty is compatible with C2, which is the intended reading. No violations.
D	Does every two-reader claim operate on fibers before scalar reduction?	4 claims, all state fiber-level interaction before reading. No violations.
E	Is every empirical claim graded and runnable?	50 inline grades: 6 THEOREM, 7 COMPUTED, 7 MEASURED, 18 DERIVED, 3 INTERPRETATION, 2 FALSIFIED, 7 OPEN.

A sixth check confirmed that no material from the header-relation branch appears anywhere in this paper. That branch is under separate investigation, has not been benched mechanically, and produced output positions inconsistent with its stated target when examined. It is excluded on the same standard the rest of the paper is held to, and its exclusion is not a judgment on it — only on its readiness to be cited.

The audit found no substantive violations. This is a weaker statement than it appears: the checks test whether the paper obeys its own stated distinctions, not whether those distinctions are correct. Section 9 supplies the falsifiers for the distinctions themselves.